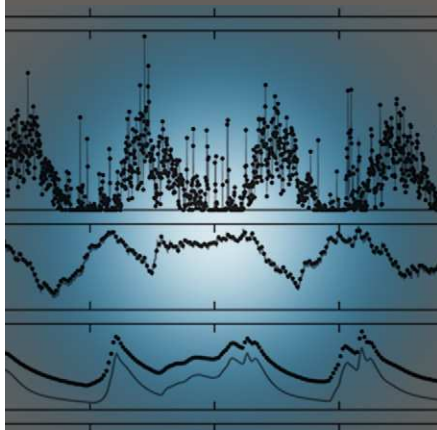


**Special Section: Frontiers
of Hydropedology in Vadose
Zone Research**

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Small-scale soil heterogeneity has a minor impact on the annual water balance as compared to ET partitioning and root water uptake with compensation. In flat terrain, one-dimensional effective models reproduce soil water budgets well, provided that the vertical series of textures in the root zones are represented well.

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Combined Impact of Soil Heterogeneity and Vegetation Type on the Annual Water Balance at the Field Scale

The hydraulic behavior of soil is determined by its hydraulic properties and their variability in space. In agricultural soils, this heterogeneity may stem from tillage or may have natural origin. The root distribution of plants will adapt to some extent to this soil heterogeneity. However, the combined impact of soil heterogeneity and root water uptake (RWU) on long-term soil water budgets has not received much attention. Numerical experiments helped identify how soil heterogeneity affects plant transpiration, soil evaporation, and groundwater recharge. Two-dimensional virtual soils with hierarchical heterogeneity, both natural and tillage induced, served as a basis for modeling soil water dynamics for a 10-yr climate record from two weather stations in Germany that vastly differ in annual precipitation. The complex interactions between soil and vegetation were explored by (i) comparing different RWU strategies (depth-, structure-, and time-dependent root profiles), (ii) land use types (perennial grass and annual winter crops), (iii) a combination of textures (silt above sand and sand above loam), and (iv) RWU with or without a compensation mechanism. The simulations were repeated with one-dimensional, effective representations of these virtual soils. In the framework of hydropedology, this study shed some light on the interaction between plants and pedological features and its impact on the macroscopic soil water budget. We demonstrated that land use has a major impact on the annual water balance through the partitioning of evapotranspiration into bare soil evaporation and plant transpiration. Compensational RWU becomes important for the annual water balance when the root zone comprises contrasting materials with respect to water holding capacity. Soil heterogeneity has in fact a minor impact on long-term soil water budgets. As a consequence, the relative contribution of plant transpiration, soil evaporation, and groundwater recharge to the total soil water loss was well reproduced by simulations in one-dimensional effective soil profiles. This advocates the application of one-dimensional soil-atmosphere-vegetation transfer (SVAT) models at larger scales. These findings only hold for assumptions made in our numerical simulations including flat area without lateral flow and no macropore flow.

Abbreviations: RWU, root water uptake; SVAT, soil-atmosphere-vegetation transfer.

Soils play a crucial role in the water cycle as they convey and partition water fluxes from the atmosphere into different directions such as plant transpiration, soil evaporation, and groundwater recharge. The unsaturated zone is an open system with predominant vertical fluxes and interactions with its surroundings mainly through the upper and lower boundary. This is why one-dimensional representations lend themselves to model soil water dynamics, especially at large temporal and spatial scales at which SVAT models and global circulation models operate. Yet soils are structured at all spatial scales (Vogel and Roth, 2003; Wagenet, 1998), with spatial heterogeneity not only in vertical but all directions. Because of the nonlinearity of soil hydraulic properties, this three-dimensional heterogeneity might cause different soil water dynamics as what could be expected from simulations with reduced dimensions and some averaged hydraulic properties (Ahuja et al., 2010; Kim et al., 1997; Milly and Eagleson, 1987; Schlüter et al., 2012b; Vogel et al., 2010; Zhu et al., 2006).

Soil formation and soil hydrology are known to interact since water flow governs dislocation processes in soil. In turn, pedogenesis induces soil heterogeneity, which controls the patterns of water flow eventually. Various traits of this relationship have been investigated within the framework of hydropedology (Lin, 2012; Lin et al., 2006). In a preceding study, Schlüter et al. (2012b) demonstrated the impact of natural and tillage-induced structural features on bare soil evaporation and groundwater recharge for two-dimensional, virtual soils with

increasing structural complexity and rather simple boundary conditions. Lateral fluxes due to inclined horizon boundaries or local barriers to vertical flow were identified as the major cause for a poor reproduction of boundary fluxes by one-dimensional effective models. This was true for artificial boundary conditions of steady state infiltration and evaporation. In the present study we explored the impact of soil heterogeneity in cases of natural boundary conditions obtained from two weather stations and focused on long-term water balances rather than on short-term transient flow.

Moreover, the incorporation of RWU by plants adds even more complexity to soil water dynamics. This has not been investigated much in conjunction with soil heterogeneity, even though both aspects are known to interact. On the one hand, variability in capillary pressure due to soil heterogeneity is mitigated by plants through stress-adapted RWU (Kuhlmann et al., 2012). On the other hand, root growth adapts to soil heterogeneity, for example, by avoiding compacted areas (Tardieu, 1988; Unger and Kaspar, 1994; Wang and Smith, 2004). In principle, this is accounted for by modeling the local RWU both as a function of water stress and root density (Feddes et al., 1976; Feddes and Raats, 2004). However, in most SVAT schemes the soil is assumed to be uniform and root density is solely modeled as function of depth and texture (Bormann, 2012; Lauenroth and Bradford, 2006; Sala et al., 1988; Sivandran and Bras, 2012; Teuling et al., 2006).

Additional complexity comes into play by considering compensation mechanisms in RWU (Šimůnek and Hopmans, 2009), which is supported by experimental evidence (Green and Clothier, 1995; Huang et al., 1997). That is, because of local stress in one part of the root system, plants increase local uptake in distant roots beyond the local demand dictated by root density to maintain the overall potential transpiration demand. While physics-based RWU models evoke compensation from fundamental principles and provide valuable insights into the process itself (Couvreur et al., 2012; de Jong van Lier et al., 2008; Jarvis, 2011; Schneider et al., 2010; Schymanski et al., 2008), empirical RWU compensation models aim at reproducing the overall effect of compensatory uptake on plant transpiration (Guswa, 2005; Jarvis, 1989; Li et al., 2001; Teuling et al., 2006). For instance, if empirical RWU compensation is incorporated into a nondimensional conceptual model, the transpiration rates of a one-dimensional Richards model in uniform soil are reproduced much better (Guswa, 2005, 2012).

Another important aspect in SVAT schemes that can have an important impact on soil water budget is the partitioning of potential evapotranspiration (ET_p) into bare soil evaporation (E_p) and plant transpiration (T_p). While T_p is integrated over the entire root zone, E_p exclusively acts on the soil surface, causing a quick depletion of soil moisture as a function of atmospheric forcing and local hydraulic properties. As a consequence, structural heterogeneity at the surface has a strong effect on the spatial distribution of soil evaporation. The partitioning of ET_p is known to depend on the coverage of the soil

surface by leaves (Alfieri et al., 2012; Persaud and Khosla, 1999), which determines the radiation energy that reaches the soil surface and the shielding of the surface from the wind.

The objective of this study was to explore the combined impact of soil heterogeneity and vegetation type on the long-term annual water balance for 10-yr datasets of atmospheric forcing at two sites in Germany differing vastly in annual precipitation. In particular, we focused on two open questions:

1. If the local RWU depends on soil water heterogeneity, what is the impact on annual water balances based on heterogeneous two-dimensional models and effective one-dimensional models?
2. How does soil heterogeneity affect the impact of vegetation type on the annual water balance?

These questions were assessed by various test scenarios: (i) different combinations of soil structure and soil texture, (ii) different root distributions in time and space, and (iii) different land use managements.

Materials and Methods

Design of a Virtual Soil

The soil in this study exhibits heterogeneity at two different scales. At the scale of observation, a 5 by 5 m domain is divided into a silty cover and a sandy subsoil with embedded loam lenses (Fig. 1). This configuration may represent a typical moraine landscape with a loess cover on top of glaciofluvial deposits. The interface between the two layers forms a 1 m deep depression that may have resulted from a paleochannel. The silty cover is superimposed by tillage-induced features such as loosely aggregated seed bed at the soil surface and a compacted plow pan at the bottom of the plow horizon, which is perforated because of biological activity. Within the silty soil cover, some subscale heterogeneity was implemented. The soil structure exhibits an anisotropic, channel-like morphology that may originate from pedogenetic translocation of fines or biological processes such root growth and the activity of soil fauna. This natural structure was destroyed in the plowed layer where an isotropic heterogeneity was generated. The loam lenses and the sand matrix in the subsoil do not exhibit inherent subscale heterogeneity as structure forming processes are attenuated in that depth. This spatial setup corresponds to soil C in a preceding study of Schlüter et al. (2012b), except the soil compaction beneath wheel tracks that have been omitted in this study. Further details on the rationale of this setup and spatial attributes of each horizon as well as a detailed description of the generation process are specified therein.

The hydraulic properties of each material are listed in Table 1, where K_s is saturated conductivity, θ_s and θ_r are the saturated and residual water contents, respectively, α and n are shape parameters of the van Genuchten retention function (van Genuchten, 1980),

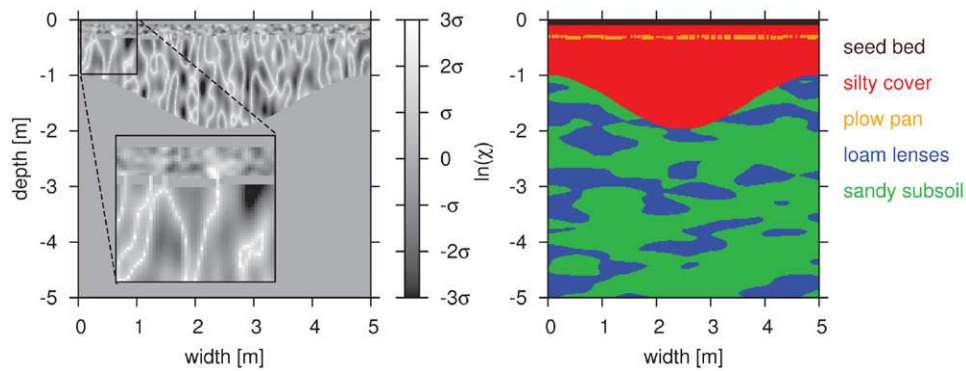


Fig. 1. Spatial distribution of subscale heterogeneity (left) and material properties (right). The inset depicts a magnification of a 1 by 1 m section.

Table 1. Mualem–van Genuchten parameters of each material in the soil model with silt over sand. Values in parenthesis for seed bed denote fractions of a bimodal retention function. Here $\sigma_{\ln(\chi)}$ is the standard deviation of scaling factors, K_s is saturated conductivity, θ_s and θ_r are the saturated and residual water contents, respectively, α and n are shape parameters of the van Genuchten retention function, and τ is tortuosity in the conductivity function.

Material	$\sigma_{\ln(\chi)}$	K_s	θ_s	θ_r	α	n	τ
		m d^{-1}			cm^{-1}		
Seed bed (0.9/0.1)	0	0.6	0.511	0.034	0.016/0.2	1.37/4	5
Silty cover	0.5	0.06	0.460	0.034	0.016	1.37	0.5
Plow pan	0	0.0006	0.379	0.039	0.0094	1.37	0.5
Sand	0	6.432	0.38	0.05	0.1	3.18	0.5
Loam	0	0.096	0.40	0.06	0.0111	1.47	0.5

and τ is tortuosity in the conductivity function (Mualem, 1976). For $n < 2$, a modified Mualem–van Genuchten parameterization was used with a fixed air-entry pressure at -2 cm (Ippisch et al., 2006, Vogel et al., 2000). Water vapor transport was considered by adding water vapor conductivity to the hydraulic conductivity function at low pressure heads. The values in Table 1 have been adapted from Schlüter et al. (2012b), except the silty cover. Here the parameter set of the silt texture class from the ROSETTA database (Schaap et al., 2001) was replaced by the corresponding set of Carsel and Parrish (1988). In this way, a more realistic unsaturated conductivity for structured silt soils was achieved. Note that some of the hydraulic parameters for the compacted plow pan and the harrowed seed bed changed accordingly. The subscale heterogeneities were implemented as Miller-similar media (Miller and Miller, 1956), each defined by a reference water retention function $h_m^*(\theta)$ and a reference conductivity function $K^*(\theta)$, where h_m (L) is capillary pressure, θ (–) is water content, and K (L/T) is hydraulic conductivity. These hydraulic reference functions correspond to a characteristic pore size r^* and the hydraulic parameters in Table 1. However, each location $\vec{x} = (x, z)$ within the subscale structure has its own microscopic length $r(\vec{x})$. The factor $\chi(\vec{x}) = r(\vec{x})/r^*$ in Fig. 1 scales both hydraulic functions according to:

$$K(\vec{x}, \theta) = K^*(\theta) \chi(\vec{x})^2 \quad [1]$$

$$h_m(\vec{x}, \theta) = h_m^*(\theta) \chi(\vec{x})^{-1} \quad [2]$$

As pore size distributions are typically log-normal, $\ln(\chi)$ may be modeled with Gaussian probability (Roth, 1995) with $\langle \ln(\chi) \rangle = 0$ and given standard deviation $\sigma_{\ln(\chi)}$. Hence, $\sigma_{\ln(\chi)} = 0$ indicates a homogeneous domain. The degree of heterogeneity $\sigma_{\ln(\chi)}$ is also listed in Table 1. Again, further details on the rationale of this hydraulic parameter set and its calculation are specified in Schlüter et al. (2012b).

The numerical experiments were performed with these settings. In addition, we implemented another virtual soil that has the same structural composition like in Fig. 1, but the textures in the subsoil are exchanged and the soil cover consists of sand instead of silt. Thus, a material with low water holding capacity and high infiltration capacity (sand) covers a material with lower infiltration capacity and higher water holding capacity (loam). In this setting no capillary barrier effect at the subsoil interface was expected. The hydraulic parameters of every material are listed in Table 2. Note that the air-entry pressure parameter α in the sandy cover was changed from 0.1 to 0.0352 like the original value for sand in ROSETTA (Schaap et al., 2001). Thus, the sandy cover is less coarse and has a higher air-entry pressure than the sand lenses in the subsoil. A seed bed was omitted as harrowing in coarse textured soils does not change the hydraulic properties of the porous material considerably. The

Table 2. Mualem–van Genuchten parameters of each material in the soil model with sand over loam. Note that there is no seed bed in the sandy cover. Here $\sigma_{\ln(\chi)}$ is the standard deviation of scaling factors, K_s is saturated conductivity, θ_s and θ_r are the saturated and residual water contents, respectively, α and n are shape parameters of the van Genuchten retention function, and τ is tortuosity in the conductivity function.

Material	$\sigma_{\ln(\chi)}$	K_s	θ_s	θ_r	α	n	τ
		m d^{-1}			cm^{-1}		
Sandy cover	0.5	6.432	0.38	0.05	0.0352	3.18	0.5
Plow pan	0	0.857	0.281	0.061	0.0206	3.18	0.5
Sand	0	6.432	0.38	0.05	0.1	3.18	0.5
Loam	0	0.096	0.40	0.06	0.0111	1.47	0.5

hydraulic parameters of the plow pan were estimated from pedo-transfer functions (Assouline, 2006a, 2006b), assuming an increase in bulk density of 15% due to compaction.

Numerical Model

The fundamental equation for unsaturated water flow at the representative elementary volume scale is the Richards equation,

$$\partial_t \theta(h_m) - \nabla \cdot [K(h_m)(\nabla h_m + z)] - U(h_m, z) = 0 \quad [3]$$

written in mixed form, where z is the vertical coordinate pointing upward and macroscopic RWU $U(h_m, z)$ is implemented as a sink term. A validity condition for Richards' equation is that h_m and θ have to be in local equilibrium even for large high hydraulic gradients that occur especially at the soil surface (Vogel and Ippisch, 2008). To achieve this, the numerical solution was implemented with an irregular grid for the numerical solution with a refinement of grid size in the upper part of the soil block. Therefore, the vertical resolution for the soil with silt covering sand increased in steps: 2 mm (0 to −0.2 m), 4 mm (−0.2 to −1 m), and 20 mm (−1 to −5 m). The horizontal resolution was 20 mm everywhere. The vertical resolution for the soil with sand covering loam was 5 mm (0 to −0.2 m), 10 mm (−0.2 to −1 m), and 62.5 mm (−1 to −5 m). In this way, computation time was reduced, which was necessary because the automatically adapted time steps were smaller due to high gradients at the soil surface every time a rain event coincided with a previously dry sandy topsoil.

The simulations were executed with the *muPhi* software package (Ippisch et al., 2006). A cell-centered finite volume scheme was used for spatial discretization. The solver used an implicit Euler scheme in time and a full upwinding scheme in space. An algebraic multigrid solver solved the linear equations. The time step was adapted automatically. Compensated RWU was implemented as an explicit Euler scheme as described below.

Root Water Uptake

Water was taken up by roots to meet the potential transpiration demand of the plant T_p (L/T). In the one-dimensional case, the uptake rate U (1/T) was distributed over the root zone as a function of capillary pressure h_m and depth z (Feddes et al., 1976):

$$U(h_m, z) = f_s(h_m) d_r(z) T_p \quad [4]$$

where $f_s(h_m)$ (–) is the root efficiency and $d_r(z)$ (1/L) is the normalized root distribution function. In the two-dimensional, structure-dependent RWU model local uptake $U(h_m, \vec{x})$ depended on the location-dependent root distribution function $d_r(\vec{x})$ (1/L²) (Šimůnek et al., 1999):

$$U(h_m, \vec{x}) = f_s(h_m) d_r(\vec{x}) L_T T_p \quad [5]$$

where L_T is the width (L) of the soil surface associated with the transpiration process.

Root Distribution

The root distribution function may have different shapes depending on the plant type, climatic conditions, stage in the growing season, and soil structural constraints. The root distribution function d_r was obtained from normalization of the local root length densities $d_r^*(\vec{x})$ (–) for the whole n -dimensional domain Ω :

$$d_r(\vec{x}) = \frac{d_r^*(\vec{x})}{\int_{\Omega} d_r^*(\vec{x}) d\Omega}$$

so that the normalized uptake rate d_r integrates to unity, that is, $\int \int d_r(\vec{x}) dx dz = 1$ for two-dimensional and $\int d_r(z) dz = 1$ for one-dimensional domains. Three different types of root distribution functions with increasing complexity are compared in Fig. 2:

1. $d_r^*(z)$: In the simplest case the root density distribution is static, independent of soil structure, and only a function of depth. This may be a valid assumption for perennial grass vegetation in vegetal equilibrium in a rather homogeneous soil. We set $d_r^*(z)$ to be a piecewise linear function (Hoffman and van Genuchten, 1983; Schelle et al., 2012):

$$d_r^*(z) = \begin{cases} 1 & : z > r_p \\ (z - r_p)/(r_{\max} - r_p) & : r_p > z > r_{\max} \\ 0 & : z < r_{\max} \end{cases} \quad [6]$$

The potential uptake rate was uniform within the plow horizon down to $r_p = -0.3$ m and linearly decreased below until $d_r(z) = 0$ was reached at the maximum root depth of $r_{\max} = -0.95$ m. This maximum root depth $r_{\max}$ was in line with the rough estimates for the latitude and potential evapotranspiration of the study sites (Breuer et al., 2003; Schenk and Jackson, 2002). A uniform root density within the plow layer and a

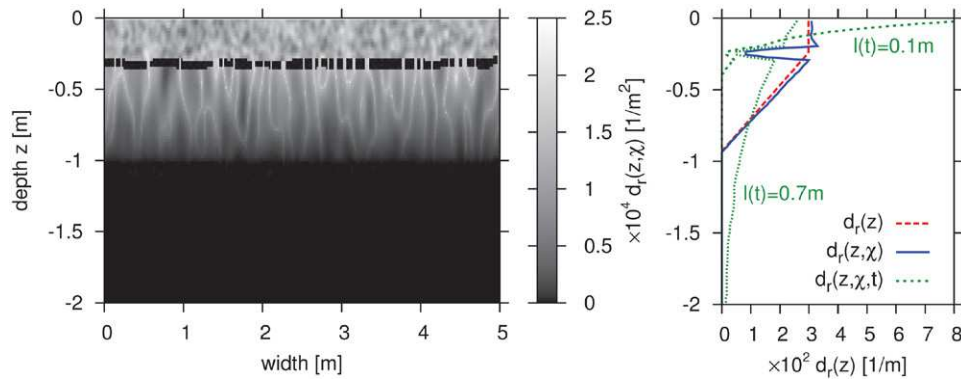


Fig. 2. Normalized two-dimensional root density distribution d_r^* as a function of depth z and scaling factor χ (left) and one-dimensional profiles of horizontally integrated d_r^* values for different distribution types (right). Note the vertical distortion in the two-dimensional image.

linear decrease below could correspond to a former agricultural field soil that was converted into pasture.

2. $d_r^*(z, \chi)$: Field studies indicate that root growth is affected by the mechanical strength of soil. That is, roots are less frequent in compacted areas (Barraclough and Weir, 1988, Ehlers et al., 1983, Tardieu, 1988, Unger and Kaspar, 1994, Wang and Smith, 2004). Moreover, fine-textured areas indicated by a low $\ln(\chi)$ value have comparatively more contacts between grains and thus exhibit a higher resistance against root penetration (Clark et al., 2003). Therefore, a structure-dependent root distribution function seemed appropriate:

$$d_r^*(z, \chi) = \begin{cases} \chi(\vec{x}) d_r^*(z) & : \vec{x} \notin \text{pan} \\ 0 & : \vec{x} \in \text{pan} \end{cases} \quad [7]$$

where $\vec{x} = (x, z)$ is the location vector and $d_r^*(z, \chi)$ is zero within the plow pan except the holes. The spatial distribution of $d_r^*(z, \chi)$ is depicted on the left-hand side of Fig. 2 together with one-dimensional averages on the right. Note the different units and ranges of d_r^* since the one-dimensional profiles represent integrated values.

3. $d_r^*(z, \chi, t)$: In contrast to perennial grass, annual crops in an agricultural field have vastly different root density distributions depending on the stage within the growing season. Root depths are shallow shortly after planting and reach a maximum depth during anthesis (Thorup-Kristensen et al., 2009). This behavior was incorporated in the following dynamic model:

$$d_r^*(z, \chi, t) = \begin{cases} \frac{\chi(\vec{x})}{l(t)} \exp\left[\frac{z}{l(t)}\right] & : \vec{x} \notin \text{pan} \\ 0 & : \vec{x} \in \text{pan} \end{cases} \quad [8]$$

Here the piecewise linear depth function was replaced by an exponential function with a time-dependent characteristic root length $l(t)$ (L). Note that $l(t)$ has a positive sign, whereas the vertical

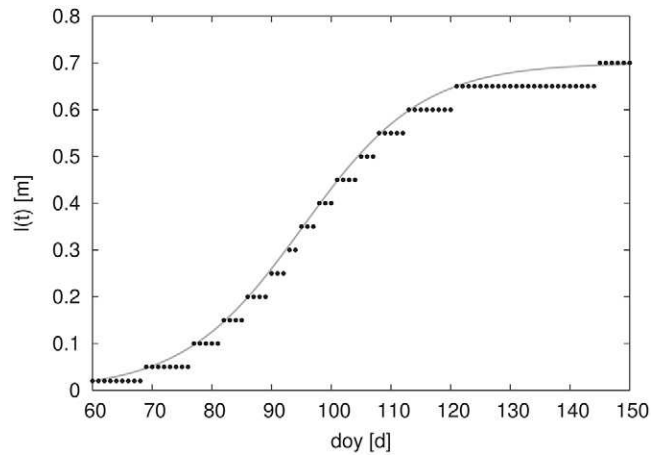


Fig. 3. Annual dynamics of characteristic root depth $l(t)$ as a function of day of year according to Eq. [8] (line) and as implemented in the numerical experiments (dots).

coordinate z has a negative sign, since z points upward and the reference height is at the soil surface. Again, the root distribution depends on soil structure, in that the scaling factors affect local root densities and roots avoid the compacted plow pan. As demonstrated in Fig. 3, shallow roots with a small initial $l_0 = 0.02$ m are assumed to be already present at the beginning of each year, as it is typical for winter crops (Thorup-Kristensen et al., 2009). At the beginning of March (Day 60), a logistic growth function (Šimůnek and Suarez, 1993) sets in that reached its maximum at $l_{\max} = 0.70$ m after 120 d according to

$$l(t) = \frac{l_{\max} l_0}{l_0 + (l_{\max} - l_0) e^{-rt}} \quad [9]$$

The growth rate r (T⁻¹) in Fig. 3 was set to 0.1 d, and the actual increase in $l(t)$ happened in regular intervals of 0.05 m to simplify its implementation in the simulation software. This type of

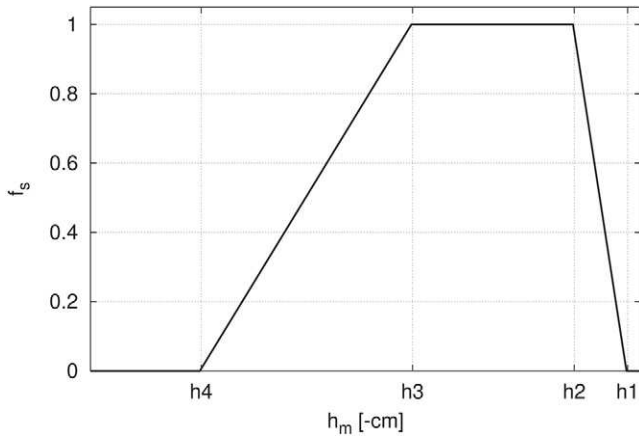


Fig. 4. Sketch of the root efficiency function with optimal root efficiency between h_2 and h_3 and no uptake below h_4 because of water stress and above h_1 because of anaerobic stress.

dynamic is justified by observed S-shaped dynamics in phenological stages (Mirschel et al., 2005) and a roughly linear relationship between aboveground biomass and root biomass (Enquist and Niklas, 2002). After harvest in August (around Day 225), we assumed a bare soil, that is, plant transpiration ceased and atmospheric water loss was routed through evaporation. For sake of simplicity, we neglected the reduction of transpiration during senescence. Also, we assumed equal dynamics for each year irrespective of temperature sums as it is typically done in plant growth models (Persaud and Khosla, 1999; Thorup-Kristensen et al., 2009).

Root Efficiency

Root water uptake by plants is reduced under unfavorable conditions, when the soil becomes too dry or too wet. This behavior was modeled with the stress function $f_s(h_m)$ (Feddes et al., 1976) that is depicted in Fig. 4. A typical set of critical capillary pressures for optimal and suboptimal conditions was used for all simulation runs with $h_1 = -10$ cm, $h_2 = -25$ cm, $h_3 = -450$ cm, and the wilting point at $h_4 = -15,000$ cm (Kuhlmann et al., 2012). The critical capillary pressure for drought stress h_3 is also a function of the transpiration rate with less negative values of h_3 for higher transpiration rates. However, this additional complexity was not considered here. Plants may overcome local stress by increasing the uptake rate at other locations in the root zone so that the total transpiration demand T_p can still be maintained. Different strategies have been suggested to incorporate this behavior into a numerical model (de Jong van Lier et al., 2008; Guswa, 2005; Jarvis, 1989; Kuhlmann et al., 2012; Li et al., 2001; Varado et al., 2006; Yadav et al., 2009). Here we introduce an empirical approach to multiply the local uptake rate by an enhancement factor M^t :

$$T_a(t) = \int M^t f_s(h_m) d_r(z) T_p(t) d\Omega \quad [10]$$

where M^t depends on the ratio between potential and actual transpiration in the last time step and is limited at 1 and $M_{\max}$:

$$M^0 = 1$$

$$M^t = \max \left[1, \min \left(M_{\max}, M^{t-1} \cdot \frac{T_p^{t-1}}{T_a^{t-1}} \right) \right] \quad [11]$$

This compensation ability $M_{\max}$ is plant specific and can range from one to more than two (Guswa, 2005). For this study we set $M_{\max} = 3$.

Interception

Interception by the plant canopy reduces or retards the incoming precipitation at the soil surface. The leaf area of plants is a key factor for interception and strongly depends on plant type, growth stage, and population density. Therefore, it is important to model interception as a dynamic, land use-specific process to reduce errors in the atmospheric forcing of the soil model. This was achieved by modeling the maximum interception storage $SI_{\max}$ as a function of leaf area index and the interception rate PI as a function of precipitation, maximum interception storage, and soil cover fraction γ (Šimůnek et al., 2008; van Dam et al., 1997):

$$SI_{\max} = \beta LAI \quad [12]$$

$$\gamma = 1 - e^{-\kappa LAI} \quad [13]$$

$$PI = SI_{\max} \left(1 - \frac{1}{1 + \frac{\gamma P}{SI_{\max}}} \right) \quad [14]$$

where LAI is leaf area index, P is gross precipitation, β is a plant-specific storage parameter typically set to 0.25 mm d^{-1} for crops (Hoyningen-Huene, 1983), and κ is the extinction coefficient set to 0.6 (Ritchie, 1972). Only the throughfall $P' = P - PI$ was applied to the soil surface. The actual storage SI was increased by PI but only until $SI_{\max}$ was reached, in which case the interception excess was added to the throughfall. Potential evapotranspiration ET_p was first applied to the canopy to deplete SI, and the rest was applied to the soil as ET' only if $ET_p > SI$.

Soil Water Balance

The soil water balance (Fig. 5) distinguishes between potential evaporation from the soil surface E' and plant transpiration from the root zone T' . Thus, the partitioned water balance is

$$I + E_a + T_a + \Delta S + G = 0$$

where I is infiltration, E_a is actual soil evaporation, T_a is actual transpiration, G is groundwater recharge, and ΔS is the water

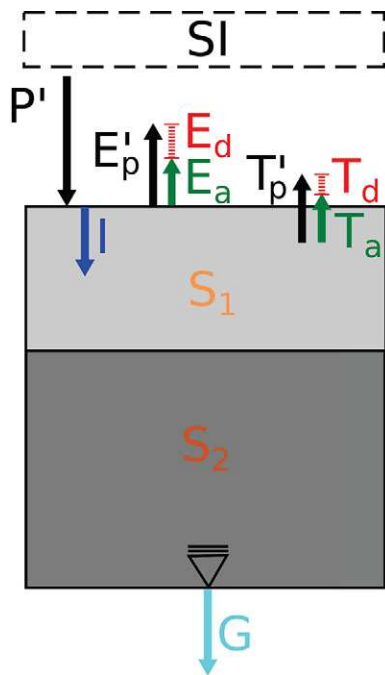


Fig. 5. Chart of all compartments of the soil water balance in which evapotranspiration is partitioned into bare soil evaporation and plant transpiration. The subscripts p, a, and d denote potential, actual, and deficit, respectively.

storage change in the soil profile. By convention, negative numbers were assigned to fluxes that depleted the storage. The storage term was partitioned into S_1 and S_2 for the soil cover and the subsoil, respectively, since they vastly differed in hydraulic behavior. The net flux at the upper boundary $P' + E'$ can be either negative for days with less rainfall than evaporation so that I vanishes or positive for days with more precipitation than evaporation so that the actual evaporation from the soil surface E_a vanishes. The difference between potential and actual evaporation or transpiration was reported separately as deficits E_d and T_d .

Initial and Boundary Conditions

A decade of atmospheric forcing was used to run the numerical experiment. The datasets stemmed from a weather station in the Jülicher Börde region in western Germany and in the Magdeburger Börde region in eastern Germany. The datasets comprised daily values for P and ET_p , where ET_p was calculated from atmospheric state variables and radiation with the Penman–Monteith equation according to the FAO guidelines (Allen et al., 1998). The ET_p was additionally partitioned into E_p and T_p on the basis of the leaf area index (LAI) according to Ritchie (1972):

$$E_p = ET_p e^{-\kappa LAI} \quad [15]$$

where κ is an extinction coefficient that was chosen to be 0.6. If infiltration excess occurred during heavy rainstorms, a temporary

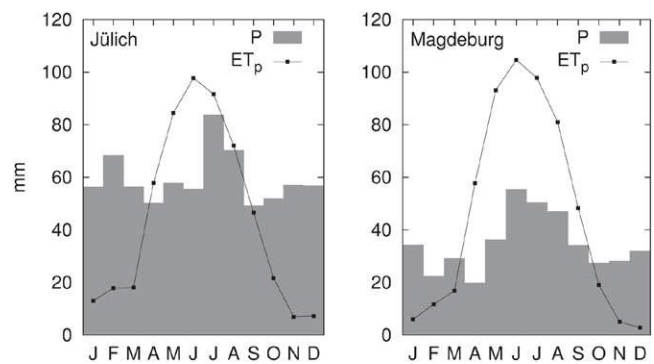


Fig. 6. Monthly averages of precipitation P and potential evapotranspiration ET_p at the Jülich site (left) and the Magdeburg site (right).

soil surface storage was filled that released the excess water subsequently. If enduring soil evaporation decreased pressure head at the upper boundary down to a limit of -10^{-5} cm, the flux boundary condition was switched to a pressure head boundary condition so that E_a diverged from E_p . The lower boundary condition was always a constant head of $h_m = 0$ cm to mimic a constant groundwater table.

As can be seen from the climate charts in Fig. 6, the Magdeburg site received less precipitation and had a more pronounced summer peak than the Jülich site because of a regional shift from maritime to continental climate. In Jülich, the mean annual precipitation of 859 mm exceeded the evapotranspiration demand of 672 mm, whereas in Magdeburg it was less with 521 mm and 653 mm, respectively.

Starting from hydraulic equilibrium, 2 yr of site-specific atmospheric forcing and model-specific soil and vegetation type served as a warm-up period so that typical soil water storages during winter served as the initial condition for each numerical experiment. Then, the simulations ran for 10 yr starting again from the first year of the climate record.

Upscaling

Various approaches exist to derive effective hydraulic properties for a given heterogeneous domain (Ahuja et al., 2010; Kim et al., 1997; Milly and Eagleson, 1987; Vogel et al., 2010; Zhu et al., 2006). Here the focus was on replacing the vertical sequence of heterogeneous, two-dimensional horizons by a one-dimensional representation with an identical sequence of horizons each possessing a single parameter set of effective hydraulic parameters. The effective water retention function was obtained from superposition of local retention functions and a new van Genuchten model fitted to the data, while the effective hydraulic conductivity function was obtained from numerical simulations and a fitted spline function. More details can be found in Schlüter et al. (2012b).

Results

The combination of different (i) climate data, (ii) soil type, (iii) vegetation type, (iv) RWU compensation, (v) root models, and (vi) dimensionality resulted in a multitude of different simulations runs. Therefore, we use a simple coding scheme in Table 3 and Table 4 to easily identify the combination used for a specific scenario. For instance, using the order of numbering above “JSiGUS1” represents the Jülich climate data (J), soil with a silty soil cover above a sandy subsoil (Si), grass as a vegetation cover (G) with uncompensated RWU (U), and a structure-dependent root model (S) in one-dimensional space (1).

Comparison between the Two Sites

Figure 7 (top) depicts the average pressure head $\langle h_m \rangle$, that is, the 10-yr arithmetic mean of θ -weighted horizontal averages, as a function of depth and day of the year (doy) for the two-dimensional soil with a structure dependent root density profile $d_r(z, \chi)$ (JSiGUS2 and MSiGUS2). The θ -weighted pressure heads were chosen because they correspond well to pressure heads in an upscaled soil with effective hydraulic properties (de Rooij, 2009; Schlüter et al., 2012a). The color range is bounded at the limits of root efficiency b_1 and b_4 . Note that in a depth of -1 m to -2 m, averaging across the wavy subsoil interface occurred, which concealed capillary barrier effects by a gradual change in capillary pressures. The difference in atmospheric forcing between Jülich (left) and Magdeburg (right) clearly resulted in different soil states throughout the entire annual cycle. In Magdeburg (right), a severe decay in root efficiency due to pressure heads close to the wilting point b_4 already starts in early June and lasts until November. Water stress was most severe directly beneath the plow horizon because during summer root density there was high but infiltrating water from rain events already taken up above. In Jülich (left), the period of reduced root efficiency is shorter and less severe. Figure 7 (middle) depicts the annual dynamics of variation coefficients of pressure heads $CV(h_m)$ within a given depth, indicating horizontal variability in pressure heads due to heterogeneous RWU or infiltration. Generally, $CV(h_m)$ was highest in a medium pressure range where the impact of local scaling factors on hydraulic functions was highest. At very high and low pressure heads, $CV(h_m)$ decreased as the all scaled hydraulic functions converged. The spatiotemporal pattern of capillary pressures was comparable for the static, structure-independent $d_r(z)$ model (JSiGUS2 and MSiGUS2) but quite different for the dynamic

Table 3. Annual water balance for a grass site with uncompensated root water uptake (RWU) and different atmospheric forcing, soil type, root model, and dimensionality. The storages S_1 and S_2 represent annual averages, whereas the fluxes PL , I , E_a , E_d , T_a , T_d , G , and ΔS represent annual sums (and proportion of annual precipitation P).

Code	Site	Soil	Land use	RWU	Root model	Dimensions	PL	I	E_a	E_d	T_a	T_d	G	ΔS	S_1	S_2
mm yr ⁻¹ /×P																
JSiGUS2	Jülich (J)	silt over sand (Si)	grass (G)	uncompensated (U)	depth (Z)	2D	112/.13	746/.87	30/.03	02/.0	517/.60	11/.01	199/.23	0/.0	51	60
JSiGUS1						1D			29/.03	04/.0	520/.60	8/.01	199/.23	0/.0	50	56
JSiGUS2					structure (S)	2D			31/.04	01/.0	515/.60	13/.01	201/.23	0/.0	51	59
JSiGUS1						1D			29/.03	03/.0	520/.60	8/.01	198/.23	0/.0	50	56
JSaGUS2						2D	112/.13	746/.87	19/.02	13/.02	391/.46	137/.16	331/.39	6/.01	13	119
JSaGUS1	Magdeburg (M)	silt over sand (Si)	grass (G)	uncompensated (U)	depth (Z)	2D	84/.16	438/.84	34/.06	3/.01	397/.76	136/.26	0/0	7/.01	34	49
MSiGUS2						1D			30/.06	7/.01	401/.77	132/.25	0/0	7/.01	32	47
MSiGUS1					structure (S)	2D			34/.06	3/.01	396/.76	137/.26	0/0	8/.01	34	49
MSaGUS2						1D			30/.06	7/.01	401/.77	132/.25	0/0	7/.01	32	47
MSaGUS1						2D	84/.16	438/.84	18/.03	19/.04	292/.56	241/.46	125/.24	3/.01	12	115
MSaGUS2	Jülich (J)	silt over sand (Si)	grass (G)	uncompensated (U)	depth (Z)	2D			14/.03	22/.04	288/.55	245/.47	132/.25	4/.01	11	88
MSaGUS1						1D			17/.03	19/.04	290/.56	243/.47	127/.24	4/.01	12	115
MSaGUS2						2D			15/.03	22/.04	288/.55	245/.47	132/.25	4/.01	11	88

Table 4. Annual water balance for one-dimensional (1D) effective soils with different atmospheric forcing, soil type, land use, root water uptake (RWU) compensation, and root models. The storages S1 and S2 represent annual averages, whereas the fluxes PI , I , E_a , E_d , T_a , T_d , G , and ΔS represent annual sums (and proportion of annual precipitation P).

Code	Site	Soil	Land use	RWU	Root model	Dimensions	PI	I	E_a	E_d	T_a	T_d	G	ΔS	S1	S2
mm yr ⁻¹ / × P																
JSiGUS1	Jülich (J)	silt over sand (Si)	grass (G)	unc. (U)	structure (S)	1D	112/.13	746/.87	29/.03	03/.0	520/.60	8/.01	198/.23	0/.0	50	56
JSiGCS1				comp. (C)					29/.03	03/.0	527/.61	1/.0	191/.22	0/.0	49	56
JSiCUD1			crops (C)	unc. (U)	dynamic (D)	1D	36/.04	823/.96	225/.26	63/.07	348/.40	1/.0	242/.28	8/.01	53	56
JSiCCD1				comp. (C)					225/.26	63/.07	349/.41	0/.0	242/.28	8/.01	53	56
JSaGUS1	Jülich (J)	sand over loam (Sa)	grass (G)	unc. (U)	structure (S)	1D	112/.13	746/.87	16/.02	16/.02	388/.45	140/.16	337/.39	6/.01	13	92
JSaGCS1				comp. (C)					14/.02	18/.02	399/.46	129/.15	324/.38	8/.01	13	92
JSaCUD1			crops (C)	unc. (U)	dynamic (D)	1D	36/.04	823/.96	178/.21	111/.13	283/.33	65/.08	350/.41	11/.01	14	91
JSaCCD1				comp. (C)					176/.20	112/.13	315/.37	34/.04	321/.37	11/.01	13	90
MSiGUS1	Magdeburg (M)	silt over sand (Si)	grass (G)	unc. (U)	structure (S)	1D	84/.16	438/.84	30/.06	7/.01	401/.77	132/.25	0/.0	7/.01	32	47
MSiGCS1				comp. (C)					26/.05	11/.02	408/.78	125/.24	0/.0	4/.01	25	47
MSiCUD1			crops (C)	unc. (U)	dynamic (D)	1D	27/.05	495/.95	168/.32	94/.18	313/.60	51/.10	0/.0	13/.03	33	43
MSiCCD1				comp. (C)					160/.31	102/.20	332/.64	32/.06	0/.0	2/.0	27	40
MSaGUS1	Magdeburg (M)	sand over loam (Sa)	grass (G)	unc. (U)	structure (S)	1D	84/.16	438/.84	15/.03	22/.04	288/.55	245/.47	132/.25	4/.01	11	88
MSaGCS1				comp. (C)					12/.02	25/.05	297/.57	236/.45	122/.23	7/.01	11	88
MSaCUD1			crops (C)	unc. (U)	dynamic (D)	1D	27/.05	495/.95	129/.25	133/.26	231/.44	134/.26	130/.25	5/.01	12	87
MSaCCD1				comp. (C)					126/.24	136/.26	276/.53	89/.17	86/.16	7/.01	12	85

$d_r(z, \chi, t)$ model (JSiGUD2 and MSiGUD2) at both sites because of a shift of RWU from the top into deeper soil layers during the growing season (not shown).

Effect of Upscaling

Figure 7 (bottom) depicts annual dynamics of the depth-dependent pressure heads h_m for the one-dimensional effective soil model with homogenous horizons (JSiGUS1 and MSiGUS1). For both sites the reproduction of wetting and drying dynamics of the two-dimensional heterogeneous soils was very good. The only difference is that the capillary barrier at the subsoil interface was clearly visible, since the interface was necessarily flat. This good reproduction was corroborated by time series of boundary fluxes for the Jülich site (Fig. 8): (i) infiltration I and plant transpiration ET_a were perfectly reproduced with a one-dimensional effective soil, implying that the fluxes at the upper boundary are effectively the same; (ii) saturation in the upper layer S1 was reproduced well, except during severe droughts when in the two-dimensional model areas with low root density still remained slightly wetter than the surrounding soil; (iii) saturation in the lower layer S2 ran generally at higher values in the two-dimensional model because of isolated water pockets in the loam lenses that vanished in the composite hydraulic functions of a homogeneous soil; and (iv) groundwater recharge in two-dimensional soils set in earlier and the increase was dampened. This is because in two-dimensional soil, water is funneled at the subsoil interface, enters the sandy subsoil preferentially from the depression, and takes some time to equilibrate laterally in the subsoil, whereas in a homogeneous soil, water enters the sandy subsoil uniformly.

In spite of these differences in soil water dynamics, the annual water balances listed in Table 3 were virtually the same for two-dimensional heterogeneous soils and one-dimensional effective soils no matter what site, soil type, or root model was considered, that is, the different short-term dynamics in groundwater recharge did not affect long-term averages.

Effect of Root Distribution

The choice of a structure-independent $d_r(z)$ or structure-dependent $d_r(z, \chi)$ root model had no effect on the annual water balance. Differences due to a dynamic $d_r(z, \chi, t)$ root model are discussed later with respect to different land uses. This indifference is demonstrated for perennial grass and uncompensated RWU (Table 3) and discussed for the soil model with silt over sand first. Local differences in RWU did not translate into long-term differences in total plant transpiration neither in Jülich (JSiGUZ2 vs. JSiGUS2) nor in Magdeburg (MSiGUZ2 vs. MSiGUS2). At both locations, roughly 15% of precipitation was intercepted by the static grass cover (PI) and all throughfall infiltrates (I) because surface runoff was inhibited. No groundwater recharge (G)

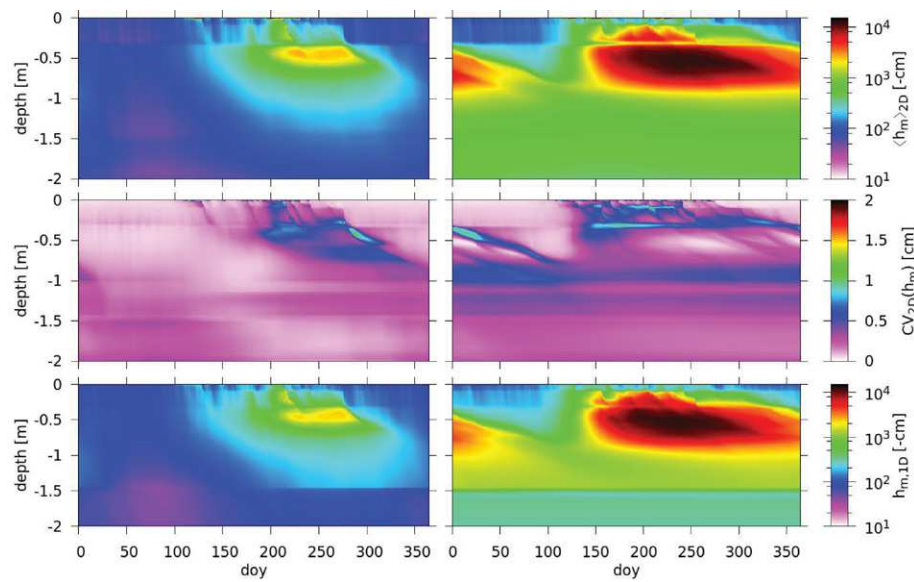


Fig. 7. (top) Profiles of average pressure heads $\langle h_m \rangle$ (10-yr arithmetic mean of θ -weighted horizontal averages) for the Jülich site (JSiGUS2, left) and the Magdeburg site (MSiGUS2, right) for the two-dimensional heterogeneous soil and a static, structure-dependent root profile $d_r(z, \chi)$. The logarithmic color range corresponds to $[h_1; h_4]$ in Fig. 6. (middle) Depth and time-dependent coefficient of variation of capillary pressure $CV(h_m)$. (bottom) Uniform pressure h_m of the one-dimensional effective soil models (JSiGUS1, MSiGUS1).

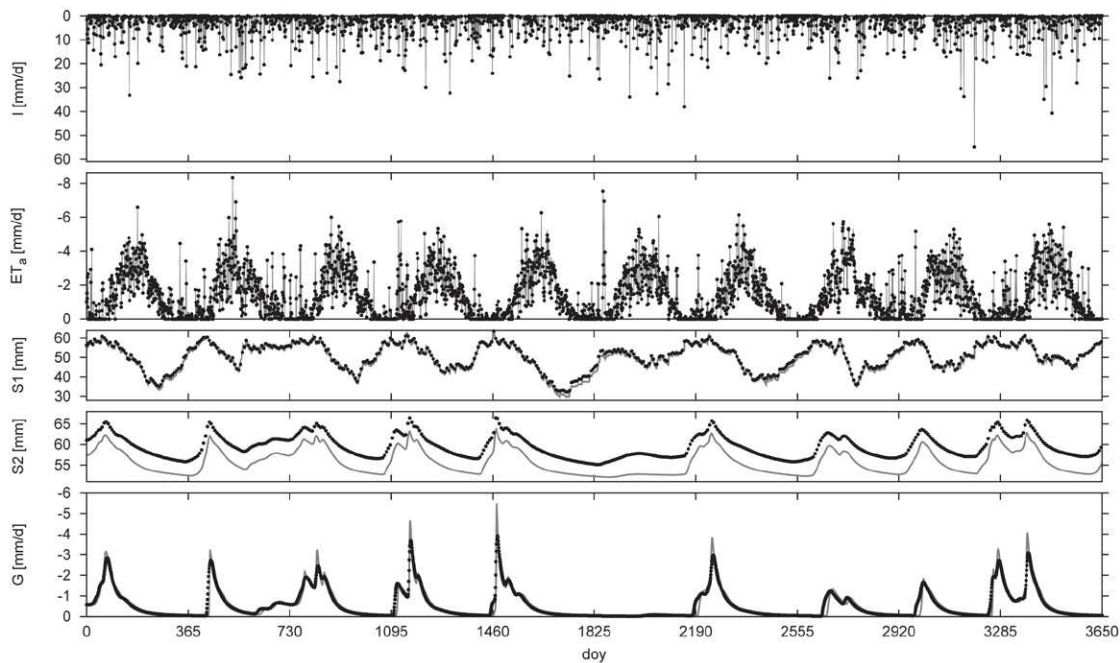


Fig. 8. Daily values of infiltration (I), evapotranspiration (ET_a), soil water storage ($S1$, $S2$), and groundwater recharge (G) for the Jülich site under grass in two-dimensional soil (JSiGUS2, black dots) and one-dimensional soil (JSiGUS1, gray).

occurred in Magdeburg so that actual evapotranspiration (ET_a) roughly equaled infiltration. In Jülich, the transpiration demand was nearly met and the infiltration surplus was translated into groundwater recharge.

Effect of Texture

An obvious explanation for the insensitivity of soil water dynamics to root distribution so far could be that the relatively high unsaturated conductivity of silt facilitated hydraulic redistribution

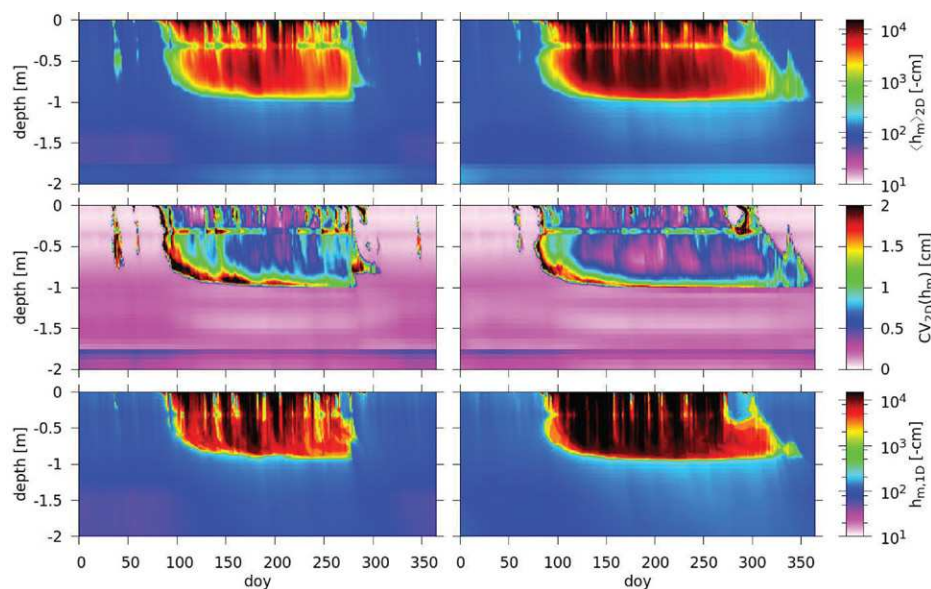


Fig. 9. Same profiles of pressure head as Fig. 7 but with exchanged textures, that is, sandy soil layer covers a loamy subsoil. (top) Averaged pressure head for JSaGUS2 (left) and MSaGUS2 (right); (middle) coefficient of variation for JSaGUS2 (left) and MSaGUS2 (right); (bottom) pressure head for JSaGUS1 (left) and MSaGUS1 (right).

through soil. In this way, pressure gradients resulting from distributed RWU were efficiently equilibrated. Therefore, a soil with exchanged textures was created where a loamy subsoil was covered by a sandy top layer. Hence, the root zone was composed of sand that exhibits low unsaturated conductivity.

Figure 9 depicts horizontal averages of capillary pressure $\langle h_m \rangle$ for the two-dimensional soil with exchanged textures (JSaGUS2 and MSaGUS2) (top), horizontal variability of capillary pressure expressed in $CV(h_m)$ (middle), and h_m for the one-dimensional effective soil with homogeneous horizons (JSaGUS1 and MSaGUS1) (bottom). The spatiotemporal patterns were quite different than the h_m profiles for the other soil type in Fig. 7:

1. The patterns for the two sites were much more similar. During winter, both sites had nearly uniform capillary pressures. The only major difference is that the period of water stress during summer and fall was more extended at the drier Magdeburg site.
2. The temporal transition from water abundance to water stress for plants when a negative net atmospheric flux set in was rather quick because of the low water holding capacity of sand.
3. Hydraulic redistribution of water through the soil from below the root zone was negligible because of low unsaturated hydraulic conductivity of sand.
4. The depth-dependent variation coefficient of pressure heads in the root zone (Fig. 9, middle) was tightly coupled to average pressure heads as discussed above. Moreover, $CV(h_m)$ peaked at higher values than in the silty root zone, since pressure head gradients cannot equilibrate due to the low unsaturated conductivity of sand.

5. The profiles of average pressure heads $\langle h_m \rangle$ were again reproduced well by the h_m profiles of the one-dimensional effective soils.

These differences in spatiotemporal patterns in capillary pressure through exchanged soil textures also had an effect on the annual water balance summarized in Table 3. Most striking was the decrease in actual transpiration of about 120 mm at both sites and the corresponding increase in groundwater recharge. Again, the difference in ET_a due to the structure dependency of the root model is negligible (JSaGUS2 vs. JSaGUS2 and MSaGUS2 vs. MSaGUS2). In this case, hydraulic redistribution does not explain this insensitivity. The reason is rather that water retention at pressure heads below the critical $h_3 = -450$ cm in the Feddes stress function is low irrespective of local scaling of hydraulic functions via χ . Thus, there is little plant-available water in the root zone that can be taken up when water stress sets in, no matter how the roots are distributed.

Since the impact of upscaling on the annual water balance is negligible for the second soil model as well, we will from here on only present results for one-dimensional effective soils and only discuss the effect of soil structural heterogeneity when it is appropriate.

Effect of Land Use

A perennial grassland site in vegetal equilibrium was compared to an agricultural site with annual winter crops. These two land use types differed in that grass was assumed to constantly cover the soil surface with $LAI = 4$ so that evapotranspiration was mainly routed through plant transpiration. The interception model also assumed a constant plant cover and the root model was static $[d_r(z, \chi)]$. In the crop site, the soil surface was only partially covered by plants

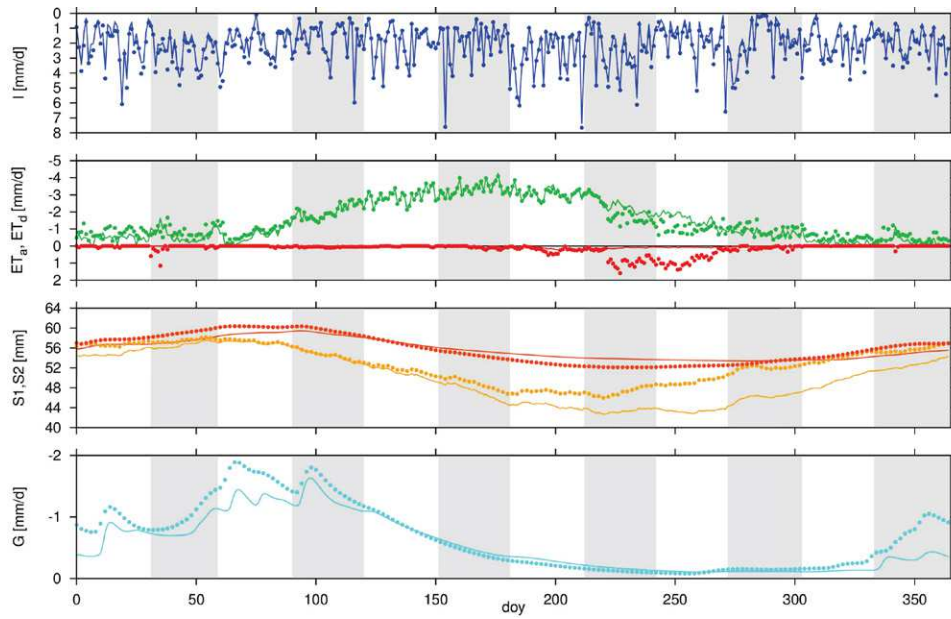


Fig. 10. Annual dynamics of the water balance terms in Jülich for grass (JSiGUS1, lines) and crops (JSiCUD1, dots). Color codes are according to the scheme in Fig. 4 (actual evapotranspiration ET_a —green; evapotranspiration deficit ET_d —red; upper soil water storage $S1$ —light orange; lower soil water storage $S2$ —dark orange).

depending on the growth stage, so that the partitioning into bare soil evaporation and plant transpiration varied over the year. The LAI dynamics that governed the interception model were part of the input data. The LAI peaked at 3.87 around the end of May. The mean root depth in the $d_r(z, \chi, t)$ root model was independent of LAI but was described by a logistic growth curve that approached its maximum around the same time.

The difference in the annual water balances of the two land use types are summarized in Table 4. Note that we first discuss results for the soil with silt over sand and uncompensated RWU ($M_{\max} = 1$). We start with the results for the Jülich site. Interception (PI) was 76 mm lower for the crop model (JSiCUD1) than for the grass model (JSiGUS1), which translated into an infiltration surplus of the same amount. In spite of more infiltration the evapotranspiration deficit ($E_d + T_d$) was 53 mm higher in the crop model. Obviously, this is because evaporation was less effective since it acted exclusively at the soil surface, whereas RWU was exerted across the entire root zone. Note that if the interception storage is filled, evapotranspiration first depletes this storage without any rate limitation and only the rest is applied to soil. Thus, $ET_p = PI + T_a + E_d + T_a + T_d$ applies. Finally, the difference in I and $E_a + T_a$ resulted in a groundwater recharge surplus of 44 mm for the crop model plus a small storage fill.

The annual dynamics of all balance terms for the Jülich site in Fig. 10 corroborates findings from the soil water budget. In addition, some seasonal effects showed up (numbering according to order of subfigures):

1. The difference in infiltration between crops and grass was highest during winter when interception was still active at the grass site, but the soil surface was bare at the crop site.
2. The highest difference in evapotranspiration deficit ($E_d + T_d$) occurred after harvest (220–260 day) when atmospheric demand was still high but entirely routed through ineffective bare soil evaporation in the crop model. In spring, evapotranspiration ($E_a + T_a$) at both sites was equal, but in winter the crop model had higher rates because all evapotranspiration was applied to the soil since a plant cover was missing, whereas at the grass site evapotranspiration was first exerted at the plant cover.
3. During summer, the crops depleted the upper soil water storage $S1$ less because some part of the transpiration demand was supplied by deep roots from the deeper soil water storage $S2$, whereas grass had a static root distribution that was restricted to the $S1$. During autumn, the difference in $S1$ was even higher but driven by Subfigure 2.
4. The difference in Subfigure 1 and Subfigure 2 translated into higher G for the crop model starting in winter (~ 310 day) and lasting until late spring (~ 120 day). In summer, however, G was slightly lower for crops with a dynamic root model because of Subfigure 3.

At the Magdeburg site, the same differences between the grass and crop site applied (MSiGUS1 vs. MSiCUD1 in Table 4). Since no groundwater recharge occurred and storage change was roughly zero on average, all infiltrating water was conveyed back to the atmosphere, the evapotranspiration deficits were equal ($E_d + T_d$), and land use could only affect timing and location. The annual dynamics of this deficit (not shown) indicate that for crops the deficit is higher

during spring because of shallow roots, lower during summer due to water uptake by deep roots, and high again after harvest due to evaporation protection by the loosely aggregated seed bed. The depletion of S_2 by deep roots during summer could not be fully recovered in winter, which resulted in lower S_2 values for the crop model.

Effect of Uptake Compensation

The numerical experiments were repeated for both sites, soils, and land use types but this time with an active compensation mechanism for stress-adapted RWU with a maximum compensation ability at $M_{\max} = 3$ summarized in Table 4. The results are discussed for the soil with silt over sand and grass first. The potential for RWU compensation in Jülich was high, since parts of the root zone were still wet when stress occurred (compare Fig. 7). However, T_d was already small for $M_{\max} = 1$ (JSiGUS1), so the increase in annual T_a of 7 mm was rather small (JSiGCS1). At the Magdeburg site, the effect of RWU compensation on the soil water balance was small, since almost all infiltrating water was used for transpiration anyway. This rather small change in absolute T_a of 7 mm due to compensational RWU was accompanied by considerable differences in transpiration dynamics expressed in terms of T_a/T_p ratios as a function of water saturation in the root zone Θ_{root} (Guswa, 2005). This is demonstrated for the grass site in Magdeburg (MSiGCS1 vs. MSaGCS1) in Fig. 11:

1. Plant transpiration at the potential rate was still maintained at lower Θ_{root} .
2. The scatter was reduced, implying that transpiration is less sensitive to vertical variability in soil moisture.
3. During severe droughts T_a/T_p for $M_{\max} = 3$ (compensation) was lower than for $M_{\max} = 1$ (no compensation) since RWU compensation leads to a more efficient depletion of Θ_{root} .

The color range depicts the annual dynamics of soil moisture and transpiration. Obviously, during late autumn, winter, and early spring Θ_{root} is high and $T_a/T_p \approx 1$. This changed during summer as demonstrated with an arbitrary trajectory for a time series from June to end of August. For $M_{\max} = 1$, T_a/T_p decreased gradually during a long drying period (155–190 day). A couple of rain events then refilled the storage up to $\Theta_{\text{root}} \approx 0.3$ again and resulted in $T_a/T_p \approx 0.6$ by the end of the examined period. The dynamics for $M_{\max} = 3$ differed in two ways: (i) instead of a gradual change during the long drying period, a high T_a/T_p ratio was maintained initially, which collapsed abruptly when the water storage in the root zone was depleted, and (ii) T_a/T_p increased immediately after a rain event since more water could be taken up directly at the soil surface. At the end of the period, T_a/T_p was at maximum again. The different dynamics were corroborated by the time series of absolute T_a values in the inset of Fig. 11. Note that the total amount of transpiration during this period was $T_a = 239$ mm for both models.

In the soil with exchanged textures (sand over loam), the effect of RWU compensation had similar effects in RWU dynamics.

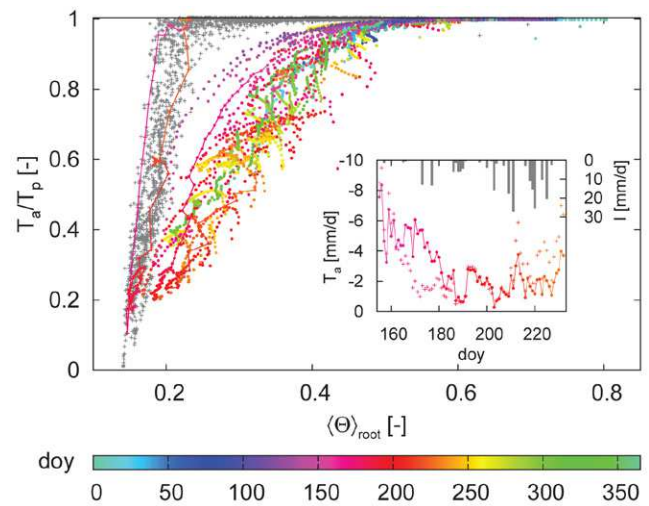


Fig. 11. Relationship between the T_a/T_p ratio and mean water saturation in the root zone $\langle \Theta_{\text{root}} \rangle$ for uncompensated RWU (MSiGUS1—colored dots) and compensated RWU (MSiGCS1—gray crosses) for Magdeburg and grass. The color code represents the annual dynamics and the trajectories depict a specific time series from June to end of August. The inset depicts the corresponding daily T_a values (color) and daily infiltration I (gray).

However, the impact on the annual water balance was slightly higher (Table 4). Since the water holding capacity in the sandy root zone was low at the grass site, a quick depletion of the storage through $M_{\max} = 3$ (JSaGCS1, MSaGCS1) made water accessible for transpiration that would otherwise be lost to groundwater recharge (JSaGUS1, MSaGUS1). The gain in actual transpiration by this accelerated water turnover was 9 to 11 mm. At the crop site, the gain in T_a by RWU compensation (JSaCCD1, MSaCCD1) was much higher (32–45 mm), because during summer the loamy S_2 with a high water holding capacity was used much more efficiently by $M_{\max} = 3$ and was refilled every year during winter. Note that upward flow at the lower boundary in -5 m depth occurred for the MSaCCD1 scenario during severe droughts because of deep RWU. However, it amounted to <1 mm in the annual water balance.

Discussion

Spatial Variability and Upscaling

We addressed the question how the combined impact of soil heterogeneity and RWU on soil water budgets changes through upscaling. In a previous study, it was demonstrated that short-term boundary fluxes during transient flow are different for one-dimensional effective soils, when lateral flow becomes relevant at the scale of observation (Schlüter et al., 2012b). This was also the case in this study concerning groundwater recharge under humid conditions (Fig. 8). For the long-term water balance, however, lateral flow was less important. Figure 12 depicts the relative contribution of different fluxes to the total annual water loss in Jülich and Magdeburg for both soil profiles and different land use types, that is, for

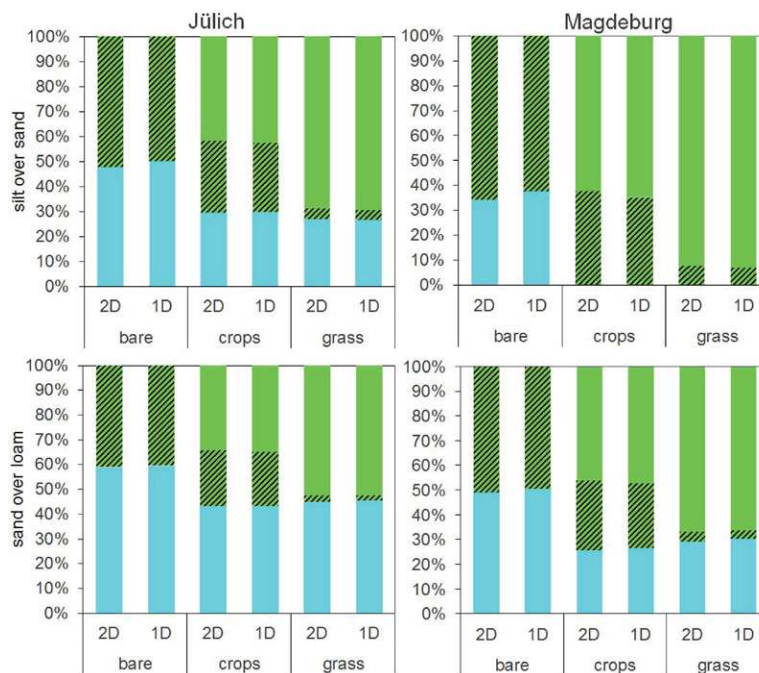


Fig. 12. Relative contribution of different fluxes to total water loss for different land use types and either a two-dimensional heterogeneous soil model or a one-dimensional effective soil model. The color codes correspond to Fig. 4 (T —green, E —hatched, G —cyan).

perennial grass, winter crops, and additionally for a bare soil, where all atmospheric demand was routed through soil evaporation. In all 12 combinations of climate, soil texture, and land use, the fractions of different water losses were roughly conserved during upscaling. This indicates that only the number, depth, and effective hydraulic behavior of a horizon and the rooting depth has to be correctly identified to reproduce the annual water balance and small-scale heterogeneity within a horizon has a minor effect.

Interplay between Soil Heterogeneity and Root Water Uptake

In some cases, this small-scale heterogeneity is of special interest and cannot be neglected even in long-term studies. For instance, the relationship between average saturation and spatial variability of saturation in soil has gained much attention in the last decades (Famiglietti et al., 1998; Hupet and Vanclooster, 2002; Ivanov et al., 2010; Lawrence and Hornberger, 2007; Pan and Peters-Lidard, 2008; Rosenbaum et al., 2012; Teuling and Troch, 2005). It is not obvious how this relationship is affected by the combined impact of soil heterogeneity and RWU. To explore this, Fig. 13 depicts the standard deviation of saturation as a function of mean saturation in 10 cm depth for grass site with a silty top layer and with structure-dependent $d_r(z, \chi)$ root model (JSiGUS2 and MSiGUS2—colored dots) or the structure-independent $d_r(z)$ root model (JSiGUZ2 and MSiGUZ2—gray circles) in Jülich (left) and Magdeburg (right). Both root models evoked a convex parabolic $\sigma(\Theta)/\langle\Theta\rangle$ relationship that is well known from numerical experiments (Lawrence and Hornberger, 2007; Teuling and Troch, 2005) and field studies

(Rosenbaum et al., 2012). Yet for the structure-dependent $d_r(z, \chi)$ root model this relationship was more scattered and less symmetric. This is because during dry-down periods some moist patches remained in areas with small local scaling factors (lower χ represents smaller pore size), because at a given pressure water retention is higher but root density and hence potential for RWU is lower. These moist patches in an otherwise dry soil caused higher $\sigma(\Theta)$ values during dry-down events. In a subsequent rain event the infiltration front advanced faster in those patches, causing temporarily high $\sigma(\Theta)$ values due to heterogeneous infiltration indicated by some outliers in Fig. 13. Interestingly, the shape of the $\sigma(\Theta)/\langle\Theta\rangle$ relationship was not affected by climate. Also, the shape of the $\sigma(\Theta)/\langle\Theta\rangle$ relationship was not affected by RWU compensation (not shown).

We demonstrated that apart from spatial variability of soil moisture, a detailed root distribution model following local bulk density is not important [$d_r(z)$ vs. $d_r(z, \chi)$] because gradients in capillary pressure from locally different RWU are quickly equilibrated via hydraulic redistribution due to high unsaturated conductivity of silt (Teuling et al., 2006). The same has been reported for heterogeneous RWU in a homogeneous loamy sand (Ivanov et al., 2010). Also, for the soil with a sandy root zone, the difference in soil water budgets through structure dependency of the root model is negligible even though the potential for hydraulic redistribution is low. This is because T_a rates are independent from the root model under no-stress conditions. During water stress, that is, $h_m < h_3$, T_a collapsed first for the $d_r(z, \chi)$ root model and then for the $d_r(z)$ root model, while the order changed with proceeding drying. Yet

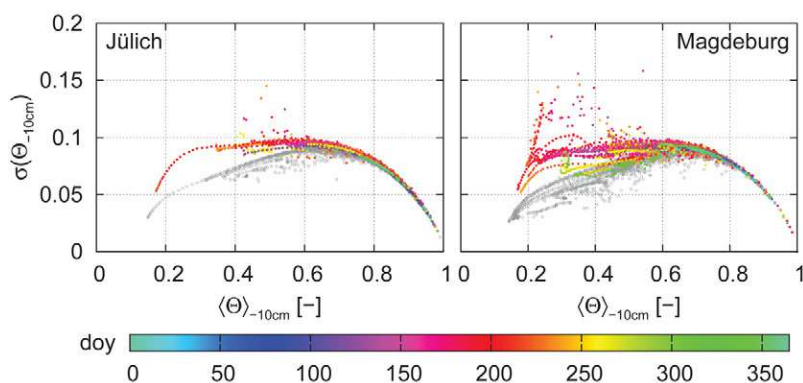


Fig. 13. Standard deviation of water saturation in -10 cm depth $\sigma(\Theta_{-10\text{cm}})$ as a function of mean saturation $\langle\Theta_{-10\text{cm}}\rangle$ for the two-dimensional heterogeneous soil under grass in Jülich (left) and Magdeburg (right). Colored dots represent the structure-dependent root model $[d_r(z, \chi) - \text{JSiGUS2, MSiGUS2}]$ and gray circles the structure-independent root model $[d_r(z) - \text{JSiGUZ2, MSiGUZ2}]$. Color code represents day of year (doy).

the differences were rather small and roughly canceled out during a dry-down event because there was little plant-available water left at $h_3 = -450$ cm because of the low water holding capacity of sand.

In turn, land use has a major impact on the soil water budget. This has been demonstrated by a comparison between annual winter crops and continuous grass vegetation. First, land use governs how precipitation is intercepted by the plant cover. This is especially important during autumn and winter when the soil at the crop site is bare and infiltration is much higher than at the grass site. Second, we could show that the partitioning of evapotranspiration into soil evaporation E and plant transpiration T is essential when modeling soil water budget under different land use. While evaporation strongly depends on soil heterogeneity at the surface, transpiration alleviates the impact of soil heterogeneity by spatially distributed RWU. As a consequence, routing of atmospheric demand through E is less effective and causes the root zone to remain at relatively high saturation even after long drying periods, which alters the initial conditions for subsequent rain events. Finally, the annual planting and harvesting at the crop site introduces seasonality in the root distribution that is less likely at the grass site.

Our results corroborated previous findings that different root distribution patterns in time are critical for water and nutrient loss to groundwater recharge in case of water abundance (Thorup-Kristensen et al., 2009) and critical for RWU from deep roots in water-limited environments (Lai and Katul, 2000; Schymanski et al., 2008). In our exercise, deep RWU resulted in an increase of annual transpiration by roughly 50 mm or 9% of annual precipitation if (i) compensated RWU was considered, which adapts to local water stress by increasing the uptake rate at other locations, (ii) transpiration demand exceeded precipitation vastly so that additional soil water storage was used to meet that demand (Magdeburg), and (iii) the vertical series of textures favored quick drainage into deeper soil layers with high water holding capacity (soil with sand over loam). In turn, if the root zone is composed of

one texture only, compensated RWU still affects timing and location of RWU including potential consequences for ET partitioning dynamics (not discussed here), which corroborates previous findings (Jarvis, 2011; Šimůnek and Hopmans, 2009; Yadav et al., 2009), but the effect on total transpiration is lower.

Limitations of the Numerical Experiments

Some properties and processes emerge under natural conditions that have been excluded from our virtual experiments. For instance, soil structural heterogeneity was assumed to be static, even though field studies demonstrate seasonal structure changes due to tillage and biological activity (Messing and Jarvis, 1993; Schlüter et al., 2011). This applies especially to macropores that are a major cause for preferential flow (Hendrickx and Flury, 2001; Jarvis, 2007). The impact on long-term water budgets, however, is hard to quantify and strongly depends on atmospheric forcing. The terrain is flat and the horizon boundaries are not inclined toward a domain boundary that inhibits lateral water fluxes that typically occur at the landscape scale as surface runoff or interflow. The implementation of small-scale heterogeneity through Miller scaling is rather simplistic. For instance, if the correlation lengths of local scaling factors would have been larger and their frequency distribution would not have been log-normal, the impact of soil structural heterogeneity on soil water budgets could have been more important.

Moreover, there is no flexibility in root growth dynamics as a function of precipitation or temperature sums as it is typically implemented in plant growth models (Persaud and Khosla, 1999; Thorup-Kristensen et al., 2009). Senescence and wilting is neglected so that the plants do not wither during a long dry period and restart to transpire immediately after the next rain event. The empirical approach to RWU compensation is without a mechanical basis and cannot reproduce common soil–plant interactions like hydraulic lift (Jarvis, 2011). Also, the coupling of soil and vegetation with the atmosphere is realized with rather simplistic, homogeneous boundary conditions. As a consequence, heterogeneous vegetation density

was not considered, which would have caused heterogeneous boundary fluxes (Alfieri et al., 2012; Teuling and Troch, 2005). However, upscaled fluxes in the water balance were somewhat insensitive to horizontal variability in soil moisture (Guswa, 2012).

Conclusions and Outlook

Despite the limitations of our numerical experiments some conclusions can be drawn with respect to the questions raised in the beginning about the combined impact of soil heterogeneity and vegetation type on the long-term soil water balance:

1. There is a dichotomy with respect to soil heterogeneity between small-scale structural heterogeneity and the vertical series of soil layers. Small-scale heterogeneity has a minor impact on RWU. Either unsaturated conductivity is sufficiently high in the pressure head range where water stress occurs, for example, silt, in which case hydraulic redistribution through soil effectively equalizes pressure head gradients caused by heterogeneous uptake, or water capacity is low in this pressure range, for example, sand, in which case the amount of plant available water was taken up by plants earlier. For these reasons, it seems to be unimportant for soil water balances in regards to whether local RWU is correlated with local hydraulic properties or not. This is the reason why in flat terrain, soil water budgets can be reproduced well by one-dimensional effective soil models despite small-scale heterogeneity that is lumped into effective parameters.
2. Land use has a major impact on soil water budget since the vegetation cover governs the partitioning into soil evaporation that is strongly affected by soil heterogeneity close to the soil surface and plant transpiration that is exerted on the root zone with a much larger support. The impact of vegetation type on the soil water balance does not depend on small-scale heterogeneity. However, if the root zone comprises contrasting materials with respect to water holding capacity, the impact of rooting strategies and RWU compensation on total plant transpiration is intensified.

From these findings we conclude that soil heterogeneity within soil horizons has a minor impact on long-term soil water budgets. This has been demonstrated for domains with small-scale variations in soil hydraulic properties, correlated heterogeneity in local RWU, and a physical model for flow (Richards' equation). This low impact of small-scale heterogeneity on soil water budgets advocates the application of one-dimensional SVAT models at larger scales provided that the vertical series of textures in the root zones is represented well. However, special care has to be taken in modeling soil–vegetation interactions like the partitioning of evapotranspiration and stress-adapted water uptake by roots. These findings only hold for the assumptions made in our numerical simulations, including a flat terrain without surface runoff or lateral flow, no macropore flow, and a simplified vegetation model.

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